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Buggy module installation: networkx & pygraphviz

I have a particular personal interest in linguistic NLP projects using Python. I found a brief project using the Princeton lexical database [WordNet](#) to create a visual graph, but there was a problem. One of the functions necessary to draw the graph produced an error that I can't resolve.

1. What did you do (Steps to reproduce)?

Here's the code from the [website](#).

```

1  from nltk.corpus import wordnet as wn
2  import networkx as nx
3  import pygraphviz
4  import matplotlib.pyplot as plt
5  def closure_graph(synset, fn):
6      seen = set()
7      graph = nx.DiGraph()
8      def recurse(s):
9          if not s in seen:
10             seen.add(s)
11             graph.add_node(s.name)
12             for s1 in fn(s):
13                 graph.add_node(s1.name)
14                 graph.add_edge(s.name, s1.name)
15                 recurse(s1)
16     recurse(synset)
17     return graph

```

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```

18 # load the dataset
19 dog = wn.synset('dog.n.01')
20 # create the graph using the custom func
21 ion closure_graph
22 graph = closure_graph(dog, lambda s: s.f
23 pernyms())
24 # plot using networkx and pygraphviz
   nx.draw_graphviz(graph)

```

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2. What happened (actual results)?

The visualization modules produce the following error:

```
AttributeError: 'module' object has no attri
```

In more detail:

```

In [94]: %run wordnet_networkx.py
-----
AttributeError Traceback (most recent call 1
/home/mastershake/xtian/Documents.x/Projects
28
29 # plot using networkx
--> 30 nx.draw_graphviz(graph)
31 #nx.draw_graphviz(graph,prog='neato')
32 #nx.draw_graphviz(graph,prog='dot')
/usr/lib/python2.7/site-packages/networkx/dr
982 See networkx.draw_networkx() for a descr
983 """
--> 984 pos = nx.drawing.graphviz_layout(G,
985 draw(G, pos, **kwargs)
986
AttributeError: 'module' object has no attri

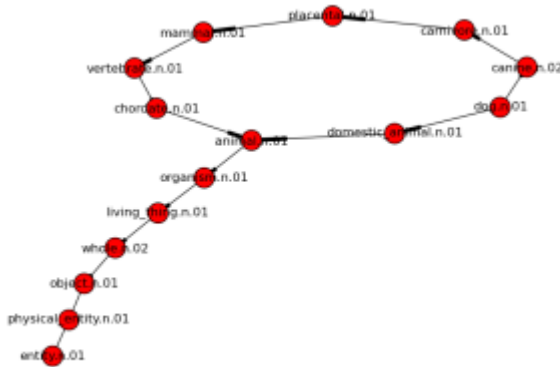
```

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3. What should have happened?

The script should have rendered a graph similar to this,



4. What did you try to fix the problem?

4.1 Verify NetworkX package requirements

[NetworkX install documentation at GitHub](#) lists package **dependencies** for NetworkX. I had everything with the exception of pyyaml. These dependencies list optional 'drawing packages', either pygraphviz or pydot, why not both!? (Really, why not?)

```
extra/python2-numpy 1.10.4-1 [installed]
community/python2-matplotlib 1.5.1-1 [instal
community/python2-scipy 0.17.0-1 [installed:
community/python2-pydot 1.0.28-3 [installed]
community/python2-pyparsing 2.0.7-1 [install
community/ipython2 4.1.1-1 [installed]
```

* (see PyDot section below, as pyparsing seems to be a dependency for PyDot with a catch.)

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I added these extra packages using a combination of Arch Linux packages, pip and easy_install. Versions of the missing packages include these,

```
In [5]: pygraphviz.__version__  
Out[5]: '1.3.1'  
In [4]: yaml.__version__  
Out[4]: '3.10'
```

So far I've not needed to install conflicting packages, which would require a more sophisticated Python module managing system (such as **Virtual Environments**)—but that might not still be the case (again, see PyDot below).

4.2 Verify PyGraphViz package requirements

I mistakenly thought PyGraphViz was a standalone Python module like Matplotlib, but reading the website reveals PyGraphViz **requires the binary package GraphViz**. Since I'm working in Arch Linux, this is easy enough with the Arch package manager.

```
extra/graphviz 2.38.0-11 [installed]  
Graph visualization software
```

The site also points to SciPy for additional dependencies. I only needed to add sympy:

```
extra/python2 2.7.11-1 [installed]  
extra/python2-nose 1.3.7-2 [installed]  
community/python2-pandas 0.17.1-2 [installed]  
community/python2-sympy 0.7.6.1.20160105-1 [
```

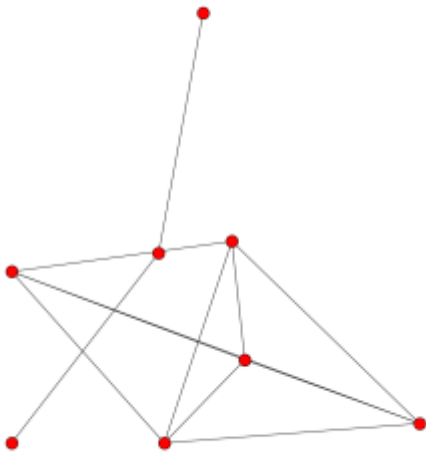
This concludes the **package requirement verifications**. Now let's continue with sample code experiments.

4.3 Test NetworkX with sample code

NetworkX [AGraph-Reference-documentation](#) lists a few example snips using Matplotlib and GraphViz.

4.3.1 NetworkX with Matplotlib–PASS!

The Matplotlib code passes and produces the following network image:



```
In [11]: import networkx as nx
In [12]: G=nx.complete_graph(5)
In [13]: G
Out[13]: <networkx.classes.graph.Graph at 0x...
In [14]: import matplotlib.pyplot as plt
In [15]: nx.draw(G)
In [16]: plt.show()
```

4.3.2 NetworkX with PyGraphViz–FAIL!

As predicted, the GraphViz snips do not work (and we already knew that!). The simplest recreation of the problem uses the same graph as the successful Matplotlib implementation.

```

In [1]: import networkx as nx
In [2]: G=nx.complete_graph(5)
In [3]: nx.draw_graphviz(G)
-----
AttributeError Traceback (most recent call 1
<ipython-input-3-481ad1c1771c> in <module>()
----> 1 nx.draw_graphviz(G)
/usr/lib/python2.7/site-packages/networkx/dr
982 See networkx.draw_networkx() for a descr
983 """
--> 984 pos = nx.drawing.graphviz_layout(G,
985 draw(G, pos, **kwargs)
986
AttributeError: 'module' object has no attri

```

Just to verify the error was not with the particular method `draw_graphviz()`, I also tested the code listed under GraphViz in the Reference. The methods `from_agraph()` and `to_agraph()` also produced errors

```

In [16]: import networkx as nx
In [17]: G=nx.complete_graph(5)
In [18]: H=nx.from_agraph(A)
-----
AttributeError Traceback (most recent call 1
<ipython-input-18-808fa68cefaa> in <module>()
----> 1 H=nx.from_agraph(A)
AttributeError: 'module' object has no attri
In [19]: A=nx.to_agraph(G)
-----
AttributeError Traceback (most recent call 1
<ipython-input-19-32d1616bb41a> in <module>()
----> 1 A=nx.to_agraph(G)
AttributeError: 'module' object has no attri
In [20]: print G
complete_graph(5)

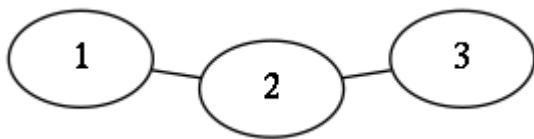
```

4.4 Test PyGraphViz with sample code–PASS!

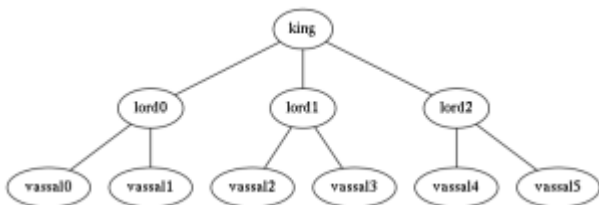
These tests suggest PyGraphViz installation is the problem. PyGraphViz's [tutorial page on Layout & Drawing](#) has some

snips as well. PyGraphViz passed with Neato (default), PyDot (see note below) and Circo Post Script output (viewed using Gimp).

```
In [1]: import pygraphviz as pgv
In [2]: d={'1': {'2': None}, '2': {'1': None}
In [3]: A=pgv.AGraph(d)
In [4]: A.write("pygraphviz_test_01.dot")
In [5]: A.layout()
In [6]: A.draw('pygraphviz_test_01.png')
```



4.5 Test PyDot with sample code–PASS!



It seems PyDot, a separate module, is also part of the mix. Reading the documentation, it was not clear to me which modules were default and therefore necessary installations (although one would think they would be listed in the dependencies). Searching the net I found a chipper coder who offers his own tests for PyDot, [Generating Graph Visualizations with pydot and Graphviz](#). In fact, I didn't have PyDot and only added the package later. During installation of PyDot I encountered strange errors. The error said I wouldn't be able to read .dot files, but still finished with OK. Fortunately, I was able to identify the problem from a StackOverflow post, [pydot and graphviz error: Couldn't import dot_parser, loading of dot files will not be possible](#). I've restarted the system and iPython several times during the process, so I feel fairly confident this did not contribute to errors; I've continually revisited the NetworkX & PyGraphViz script to check if anything has resolved the problem.

However, there is one odd duck in the PyDot install. The Arch package manager says python2-pyparsing 2.0.7-1 is the

installed version. From the above install fix, we in iPython environment we find `pyparsing.__version__` is '1.5.7'. This is what is supposed to be installed, and so far it's not a problem. Nevertheless, this raises the question whether or not the other modules are installed correctly for NetworkX and PyGraphViz.

5. Next Steps

There are no exact matches on StackOverflow, so I'm tending to think the problem is a peculiarity of my system install. The problem seems to rest with pygraphviz. I will first try a post at [StackOverflow](#). If that doesn't work, I will try to open a bug ticket to get this case reviewed and possibly improve the documentation. Hopefully the maintainers will have an insight as to what the error signifies. Why do some features work while others do not? Also, the tutorial is rather old (2010) so the projects could have made changes, causing the sample code to be obsolete—not that there's much to the example. But I'm just coming to the module for the first time, and don't know what's the correct implementation yet.

Update

I posted to StackOverflow, [What could cause NetworkX & PyGraphViz to work fine alone but not together?](#) and shortly received the answer from member Aric:

”

There is a small bug in the `draw_graphviz` function in `networkx-1.11` triggered by the change that the `graphviz` drawing tools are no longer imported into the top level namespace of `networkx`.

“

Footnote

Working through the example code, [“intializing a graph ... using a string”](#), for PyGraphViz there might be another bug

or error in the documentation. Below I work over the sample code:

```
In [25]: import pygraphviz as pgv
In [26]: G=pgv.AGraph()
In [27]: G=pgv.AGraph('graph {1 - 2}')
Error: syntax error in line 1 near '-'
-----
DotError Traceback (most recent call last)
<ipython-input-27-e7d3bd18b295> in <module>()
----> 1 G=pgv.AGraph('graph {1 - 2}')
[...]
```

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